PRACTICAL 2

Hadoop Installation, Configuration and HDFS Operations

Subject: Big Data Analytics (BDA)

1. Aim

The aim of this practical was to check the Hadoop installation and perform basic HDFS operations. The work included creating HDFS directories, uploading a file, viewing its contents, copying, moving, renaming, downloading and deleting files, and finally checking the health and storage status of HDFS.

2. Software and Environment Used

The practical was performed in an Ubuntu/WSL Linux environment using the terminal and Bash shell. The setup already had OpenJDK 11.0.31 and Hadoop 3.4.1 installed. A single-node Hadoop cluster was used for the HDFS operations.

3. Test Data Used

A small sales data file named sales_data.txt was used so that the HDFS commands could be tested easily. The file contained the following sample records:

ProductA 120
ProductB 80
ProductC 150
ProductA 200
ProductB 90

4. Creating the Working Folder

Before starting the HDFS operations, I created a separate local folder for this practical. This made it easier to keep the sample file and downloaded file in one place.

mkdir -p ~/Practicals/Practical_02_Hadoop_Installation_and_HDFS

cd ~/Practicals/Practical_02_Hadoop_Installation_and_HDFS

pwd

ls -la

```
hit@Hit:~$ mkdir -p ~/Practicals/Practical_02_Hadoop_Installation_and_HDFS
hit@Hit:~$ cd ~/Practicals/Practical_02_Hadoop_Installation_and_HDFS
hit@Hit:~/Practicals/Practical_02_Hadoop_Installation_and_HDFS$ pwd
ls -la
/home/hit/Practicals/Practical_02_Hadoop_Installation_and_HDFS
total 8
drwxr-xr-x 2 hit hit 4096 Sep 11 17:14 .
drwxr-xr-x 5 hit hit 4096 Sep 11 17:14 ..
hit@Hit:~/Practicals/Practical_02_Hadoop_Installation_and_HDFS$
```

5. Checking Java and Hadoop Installation

Next, I checked whether Java and Hadoop were available correctly. I also checked the Hadoop and Java environment variables and opened the Hadoop configuration folder.

java -version

hadoop version

echo $HADOOP_HOME

echo $JAVA_HOME

cd $HADOOP_HOME/etc/hadoop

ls

```
hit@Hit:~/Practicals/Practical_02_Hadoop_Installation_and_HDFS$ java -version
openjdk version "11.0.31" 2026-04-21
OpenJDK Runtime Environment (build 11.0.31+11-post-1ubuntu1-26.04.2-Ubuntu)
OpenJDK 64-Bit Server VM (build 11.0.31+11-post-1ubuntu1-26.04.2-Ubuntu, mixed mode, sharing)
hit@Hit:~/Practicals/Practical_02_Hadoop_Installation_and_HDFS$ hadoop version
Hadoop 3.4.1
Source code repository https://github.com/apache/hadoop.git -r 4d7825309348956336b8f06a08322b78422849b1
Compiled by mthakur on 2024-10-09T14:57Z
Compiled on platform linux-x86_64
Compiled with protoc 3.23.4
From source with checksum 7292fe9dba5e2e44e3a9f763fce3e680
This command was run using /home/hit/hadoop-3.4.1/share/hadoop/common/hadoop-common-3.4.1.jar
hit@Hit:~/Practicals/Practical_02_Hadoop_Installation_and_HDFS$ echo $HADOOP_HOME
echo $JAVA_HOME
/home/hit/hadoop-3.4.1
/usr/lib/jvm/java-11-openjdk-amd64
hit@Hit:~/Practicals/Practical_02_Hadoop_Installation_and_HDFS$ cd $HADOOP_HOME/etc/hadoop
hit@Hit:~/hadoop-3.4.1/etc/hadoop$ cd $HADOOP_HOME/etc/hadoop
hit@Hit:~/hadoop-3.4.1/etc/hadoop$ ls
capacity-scheduler.xml      hadoop-policy.xml                kms-acls.xml          mapred-queues.xml.template     yarn-env.cmd
configuration.xsl           hadoop-user-functions.sh.example kms-env.sh            mapred-site.xml                yarn-env.sh
container-executor.cfg      hdfs-rbf-site.xml                kms-log4j.properties  shellprofile.d                 yarn-site.xml
core-site.xml               hdfs-site.xml                    kms-site.xml          ssl-client.xml.example         yarnservice-log4j.properties
hadoop-env.cmd              httpfs-env.sh                    log4j.properties      ssl-server.xml.example
hadoop-env.sh               httpfs-log4j.properties          mapred-env.cmd        user_ec_policies.xml.template
hadoop-metrics2.properties  httpfs-site.xml                  mapred-env.sh         workers
hit@Hit:~/hadoop-3.4.1/etc/hadoop$
```

6. Checking Hadoop Configuration

The main configuration files were checked to confirm the current single-node setup. I did not change the files at this stage; I only verified the values already configured.

6.1 core-site.xml

cat core-site.xml

The file showed fs.defaultFS as hdfs://localhost:9000, which is the HDFS address used in this setup.

```
hit@Hit:~/hadoop-3.4.1/etc/hadoop$ cat core-site.xml
<?xml version="1.0" encoding="UTF-8"?>
<configuration>

<property>
    <name>fs.defaultFS</name>
    <value>hdfs://localhost:9000</value>
</property>

</configuration>
```

6.2 hdfs-site.xml

cat hdfs-site.xml

The replication factor was set to 1 because this practical used one DataNode. The NameNode and DataNode storage locations were also visible in the file.

```
hit@Hit:~/hadoop-3.4.1/etc/hadoop$ cat hdfs-site.xml
<?xml version="1.0" encoding="UTF-8"?>
<configuration>

<property>
    <name>dfs.replication</name>
    <value>1</value>
</property>

<property>
    <name>dfs.namenode.name.dir</name>
    <value>file:///home/hit/hadoopdata/hdfs/namenode</value>
</property>

<property>
    <name>dfs.datanode.data.dir</name>
    <value>file:///home/hit/hadoopdata/hdfs/datanode</value>
</property>

</configuration>
```

6.3 yarn-site.xml

cat yarn-site.xml

The YARN configuration file was also checked as part of verifying the Hadoop configuration.

```
hit@Hit:~/hadoop-3.4.1/etc/hadoop$ cat yarn-site.xml
<?xml version="1.0"?>
<!--
  Licensed under the Apache License, Version 2.0 (the "License");
  you may not use this file except in compliance with the License.
  You may obtain a copy of the License at

    http://www.apache.org/licenses/LICENSE-2.0

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  distributed under the License is distributed on an "AS IS" BASIS,
  WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
  See the License for the specific language governing permissions and
  limitations under the License. See accompanying LICENSE file.
-->
<configuration>

<!-- Site specific YARN configuration properties -->

</configuration>
```

7. Returning to the Practical Folder

After checking the configuration files, I returned to the local practical folder before creating the sample data file.

cd ~/Practicals/Practical_02_Hadoop_Installation_and_HDFS

pwd

```
hit@Hit:~/hadoop-3.4.1/etc/hadoop$ cd ~/Practicals/Practical_02_Hadoop_Installation_and_HDFS
pwd
/home/hit/Practicals/Practical_02_Hadoop_Installation_and_HDFS
hit@Hit:~/Practicals/Practical_02_Hadoop_Installation_and_HDFS$
```

8. Starting and Checking Hadoop Services

The Hadoop services were started using the configured start command. After that, jps was used to see whether the required Java processes were running.

start-hadoop

jps

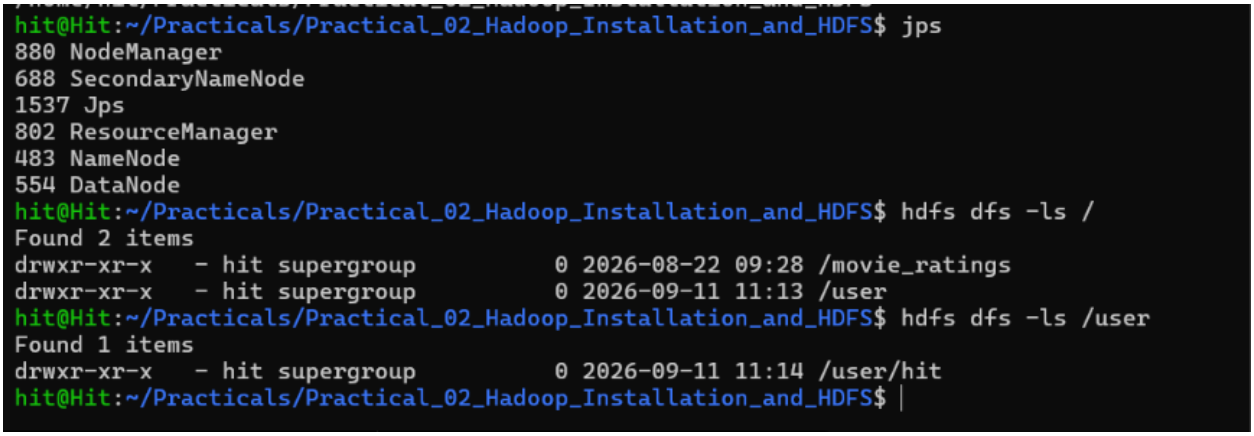
The output showed NameNode, DataNode, SecondaryNameNode, ResourceManager and NodeManager. This confirmed that the main services required for the single-node practical were running.

```
hit@Hit:~/Practicals/Practical_02_Hadoop_Installation_and_HDFS$ jps
880 NodeManager
688 SecondaryNameNode
1537 Jps
802 ResourceManager
483 NameNode
554 DataNode
hit@Hit:~/Practicals/Practical_02_Hadoop_Installation_and_HDFS$
```

9. Checking the Existing HDFS Structure

Before creating the practical folder in HDFS, I checked the root directory and the user directory.

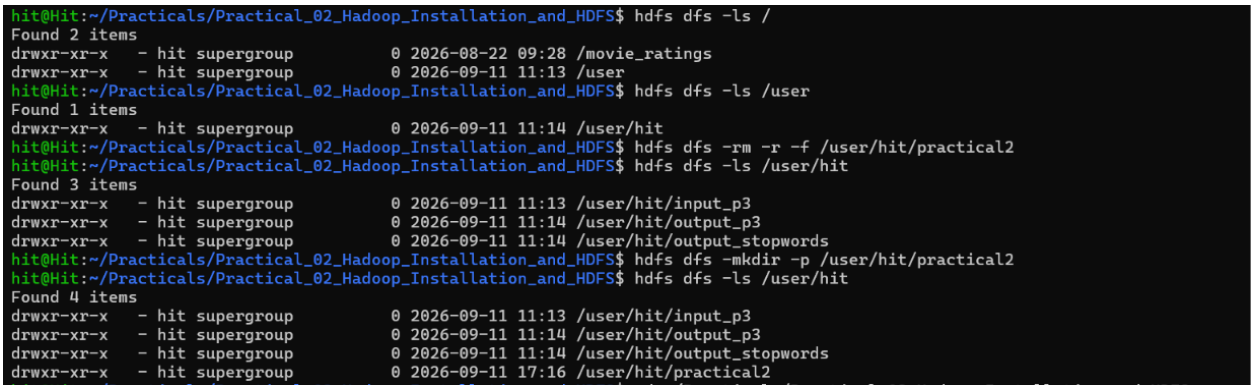
```
hdfs dfs -ls /
hdfs dfs -ls /user
```



10. Creating the HDFS Directory

For a clean run, the old practical2 directory was removed first if it existed. Then a new practical2 directory was created under /user/hit.

```
hdfs dfs -rm -r -f /user/hit/practical2
hdfs dfs -mkdir -p /user/hit/practical2
hdfs dfs -ls /user/hit
```



11. Creating and Checking the Local Sales Data File

The sample file was created locally. After creating it, I displayed the contents, checked the file size and counted the number of lines.

```
cat << 'EOF' > sales_data.txt
ProductA 120
ProductB 80
ProductC 150
ProductA 200
ProductB 90
```

EOF

cat sales_data.txt

```
hit@Hit:~/Practicals/Practical_02_Hadoop_Installation_and_HDFS$ cd ~/Practicals/Practical_02_Hadoop_Installation_and_HDFS
hit@Hit:~/Practicals/Practical_02_Hadoop_Installation_and_HDFS$ cat << 'EOF' > sales_data.txt
ProductA 120
ProductB 80
ProductC 150
ProductA 200
ProductB 90
EOF
hit@Hit:~/Practicals/Practical_02_Hadoop_Installation_and_HDFS$ cat sales_data.txt
ProductA 120
ProductB 80
ProductC 150
ProductA 200
ProductB 90
hit@Hit:~/Practicals/Practical_02_Hadoop_Installation_and_HDFS$
```

ls -lh sales_data.txt

wc -l sales_data.txt

```
hit@Hit:~/Practicals/Practical_02_Hadoop_Installation_and_HDFS$ ls -lh sales_data.txt
-rw-r--r-- 1 hit hit 63 Sep 11 17:16 sales_data.txt
hit@Hit:~/Practicals/Practical_02_Hadoop_Installation_and_HDFS$ wc -l sales_data.txt
5 sales_data.txt
hit@Hit:~/Practicals/Practical_02_Hadoop_Installation_and_HDFS$
```

12. Uploading the File to HDFS

Once the local file was ready, it was uploaded into the practical2 directory in HDFS. The directory was listed afterwards to confirm that the upload worked.

hdfs dfs -put sales_data.txt /user/hit/practical2/

hdfs dfs -ls /user/hit/practical2

```
cp: `/user/hit/practical2/sales_data.txt': No such file or directory
hit@Hit:~/Practicals/Practical_02_Hadoop_Installation_and_HDFS$ hdfs dfs -put sales_data.txt /user/hit/practical2/
hit@Hit:~/Practicals/Practical_02_Hadoop_Installation_and_HDFS$ hdfs dfs -ls /user/hit/practical2
Found 1 items
-rw-r--r--   1 hit supergroup         63 2026-09-11 17:18 /user/hit/practical2/sales_data.txt
hit@Hit:~/Practicals/Practical_02_Hadoop_Installation_and_HDFS$
```

13. Displaying the File Directly from HDFS

The cat command was used with HDFS to confirm that the uploaded file could be read correctly.

hdfs dfs -cat /user/hit/practical2/sales_data.txt

```
hit@Hit:~/Practicals/Practical_02_Hadoop_Installation_and_HDFS$ hdfs dfs -cat /user/hit/practical2/sales_data.txt
ProductA 120
ProductB 80
ProductC 150
ProductA 200
ProductB 90
hit@Hit:~/Practicals/Practical_02_Hadoop_Installation_and_HDFS$
```

14. Copying a File Inside HDFS

A copy of the sales file was created inside the same HDFS directory. The directory was listed again so both files could be seen.

hdfs dfs -cp /user/hit/practical2/sales_data.txt /user/hit/practical2/sales_backup.txt

hdfs dfs -ls /user/hit/practical2

```
hit@Hit:~/Practicals/Practical_02_Hadoop_Installation_and_HDFS$ hdfs dfs -cp /user/hit/practical2/sales_data.txt /user/hit/practical2/sales_backup.txt
hdfs dfs -ls /user/hit/practical2
Found 2 items
-rw-r--r--   1 hit supergroup         63 2026-09-11 17:18 /user/hit/practical2/sales_backup.txt
-rw-r--r--   1 hit supergroup         63 2026-09-11 17:18 /user/hit/practical2/sales_data.txt
hit@Hit:~/Practicals/Practical_02_Hadoop_Installation_and_HDFS$
```

15. Moving and Renaming the Backup File

An archive directory was created to test moving files. The backup file was moved into the archive directory and then renamed to final_sales_backup.txt.

hdfs dfs -mkdir /user/hit/practical2/archive

hdfs dfs -mv /user/hit/practical2/sales_backup.txt /user/hit/practical2/archive/

hdfs dfs -ls /user/hit/practical2/archive

hdfs dfs -mv /user/hit/practical2/archive/sales_backup.txt /user/hit/practical2/archive/final_sales_backup.txt

hdfs dfs -ls /user/hit/practical2/archive

```
hit@Hit:~/Practicals/Practical_02_Hadoop_Installation_and_HDFS$ hdfs dfs -cp /user/hit/practical2/sales_data.txt /user/hit/practical2/sales_backup.txt
hdfs dfs -ls /user/hit/practical2
Found 2 items
-rw-r--r--   1 hit supergroup         63 2026-09-11 17:18 /user/hit/practical2/sales_backup.txt
-rw-r--r--   1 hit supergroup         63 2026-09-11 17:18 /user/hit/practical2/sales_data.txt
hit@Hit:~/Practicals/Practical_02_Hadoop_Installation_and_HDFS$ hdfs dfs -put sales_data.txt /user/hit/practical2/
hdfs dfs -ls /user/hit/practical2
put: `/user/hit/practical2/sales_data.txt': File exists
Found 2 items
-rw-r--r--   1 hit supergroup         63 2026-09-11 17:18 /user/hit/practical2/sales_backup.txt
-rw-r--r--   1 hit supergroup         63 2026-09-11 17:18 /user/hit/practical2/sales_data.txt
hit@Hit:~/Practicals/Practical_02_Hadoop_Installation_and_HDFS$ hdfs dfs -cat /user/hit/practical2/sales_data.txt
ProductA 120
ProductB 80
ProductC 150
ProductA 200
ProductB 90
hit@Hit:~/Practicals/Practical_02_Hadoop_Installation_and_HDFS$
```

16. Downloading the File Back to Local Storage

To check the reverse operation, the original sales file was downloaded from HDFS into a local folder. The downloaded file was then listed and displayed.

mkdir -p downloaded_files

hdfs dfs -get /user/hit/practical2/sales_data.txt downloaded_files/

ls -lh downloaded_files/sales_data.txt

cat downloaded_files/sales_data.txt

```
hit@Hit:~/Practicals/Practical_02_Hadoop_Installation_and_HDFS$ ^[[200~hdfs dfs -cp /user/hit/practical2/sales_data.txt /user/hit/practical2/sales_backup.txt
s dfs -ls /user/hit/practical2^[[201~
hdfs: command not found
hit@Hit:~/Practicals/Practical_02_Hadoop_Installation_and_HDFS$ hdfs dfs -ls /user/hit/practical2~
ls: `/user/hit/practical2~': No such file or directory
hit@Hit:~/Practicals/Practical_02_Hadoop_Installation_and_HDFS$ hdfs dfs -mkdir /user/hit/practical2/archive
hdfs dfs -mv /user/hit/practical2/sales_backup.txt /user/hit/practical2/archive/
hdfs dfs -ls /user/hit/practical2
hdfs dfs -ls /user/hit/practical2/archive
Found 2 items
drwxr-xr-x   - hit supergroup          0 2026-09-11 17:19 /user/hit/practical2/archive
-rw-r--r--   1 hit supergroup         63 2026-09-11 17:18 /user/hit/practical2/sales_data.txt
Found 1 items
-rw-r--r--   1 hit supergroup         63 2026-09-11 17:18 /user/hit/practical2/archive/sales_backup.txt
hit@Hit:~/Practicals/Practical_02_Hadoop_Installation_and_HDFS$ hdfs dfs -mv /user/hit/practical2/archive/sales_backup.txt /user/hit/practical2/archive/final_sales_backup.txt
hdfs dfs -ls /user/hit/practical2/archive
Found 1 items
-rw-r--r--   1 hit supergroup         63 2026-09-11 17:18 /user/hit/practical2/archive/final_sales_backup.txt
hit@Hit:~/Practicals/Practical_02_Hadoop_Installation_and_HDFS$ mkdir -p downloaded_files
hdfs dfs -get /user/hit/practical2/sales_data.txt downloaded_files/
ls -lh downloaded_files
cat downloaded_files/sales_data.txt
total 4.0K
-rw-r--r-- 1 hit hit 63 Sep 11 17:20 sales_data.txt
ProductA 120
ProductB 80
ProductC 150
ProductA 200
ProductB 90
hit@Hit:~/Practicals/Practical_02_Hadoop_Installation_and_HDFS$ hdfs dfsadmin -report
Configured Capacity: 10811011176832 (1006.85 GB)
```

17. Deleting a File and an Empty Directory

After testing the move and rename operations, the temporary backup file was no longer required. It was deleted first, and then the now-empty archive directory was removed.

hdfs dfs -rm /user/hit/practical2/archive/final_sales_backup.txt

hdfs dfs -rmdir /user/hit/practical2/archive

hdfs dfs -ls /user/hit/practical2

```
hit@Hit:~/Practicals/Practical_02_Hadoop_Installation_and_HDFS$ hdfs dfs -rm /user/hit/practical2/archive/final_sales_backup.txt
hdfs dfs -ls /user/hit/practical2/archive
Deleted /user/hit/practical2/archive/final_sales_backup.txt
hit@Hit:~/Practicals/Practical_02_Hadoop_Installation_and_HDFS$ hdfs dfs -rmdir /user/hit/practical2/archive
hdfs dfs -ls /user/hit/practical2
Found 1 items
-rw-r--r--   1 hit supergroup         63 2026-09-11 17:18 /user/hit/practical2/sales_data.txt
hit@Hit:~/Practicals/Practical_02_Hadoop_Installation_and_HDFS$ hdfs dfs -ls -R /user/hit/practical2
-rw-r--r--   1 hit supergroup         63 2026-09-11 17:18 /user/hit/practical2/sales_data.txt
hit@Hit:~/Practicals/Practical_02_Hadoop_Installation_and_HDFS$
```

18. Checking HDFS Block and File Health

The fsck command was used to check the stored file. The output showed the file status as HEALTHY and also displayed its block-related information.

hdfs fsck /user/hit/practical2/sales_data.txt -files -blocks -locations

```
hit@Hit: ~/Practicals/Practica
ProductB 80
ProductC 150
ProductA 200
ProductB 90
hit@Hit:~/Practicals/Practical_02_Hadoop_Installation_and_HDFS$ hdfs dfsadmin -report
Configured Capacity: 1081101176832 (1006.85 GB)
Present Capacity: 1021415182484 (951.27 GB)
DFS Remaining: 1021415092224 (951.27 GB)
DFS Used: 90260 (88.14 KB)
DFS Used%: 0.00%
Replicated Blocks:
        Under replicated blocks: 0
        Blocks with corrupt replicas: 0
        Missing blocks: 0
        Missing blocks (with replication factor 1): 0
        Low redundancy blocks with highest priority to recover: 0
        Pending deletion blocks: 0
Erasure Coded Block Groups:
        Low redundancy block groups: 0
        Block groups with corrupt internal blocks: 0
        Missing block groups: 0
        Low redundancy blocks with highest priority to recover: 0
        Pending deletion blocks: 0

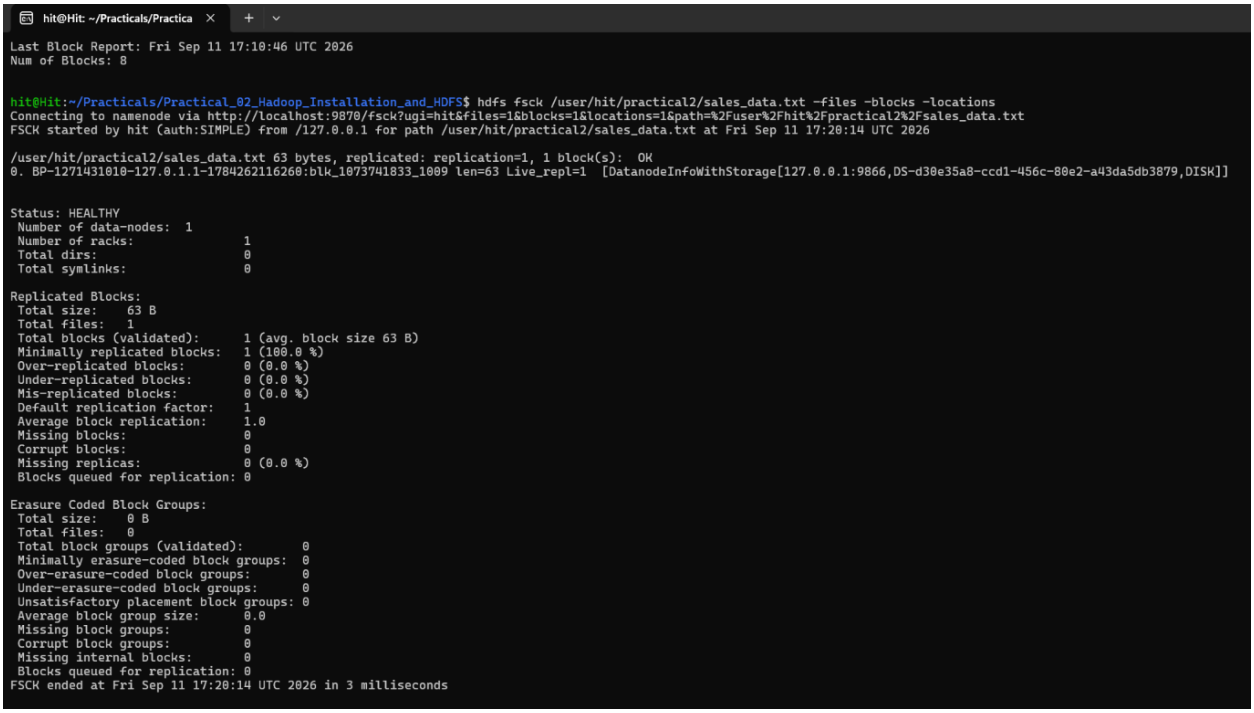
-------------------------------------------------
Live datanodes (1):

Name: 127.0.0.1:9866 (localhost)
Hostname: Hit.localdomain
Decommission Status : Normal
Configured Capacity: 1081101176832 (1006.85 GB)
DFS Used: 90260 (88.14 KB)
Non DFS Used: 4693639020 (4.37 GB)
DFS Remaining: 1021415092224 (951.27 GB)
DFS Used%: 0.00%
DFS Remaining%: 94.48%
Configured Cache Capacity: 0 (0 B)
Cache Used: 0 (0 B)
Cache Remaining: 0 (0 B)
Cache Used%: 100.00%
Cache Remaining%: 0.00%
Xceivers: 0
Last contact: Fri Sep 11 17:20:07 UTC 2026
Last Block Report: Fri Sep 11 17:10:46 UTC 2026
Num of Blocks: 8
```

19. Checking HDFS Storage and DataNode Status

The HDFS administration report was checked to see the storage details and the live DataNode information.

hdfs dfsadmin -report



20. Checking Cluster Health and Storage Utilization

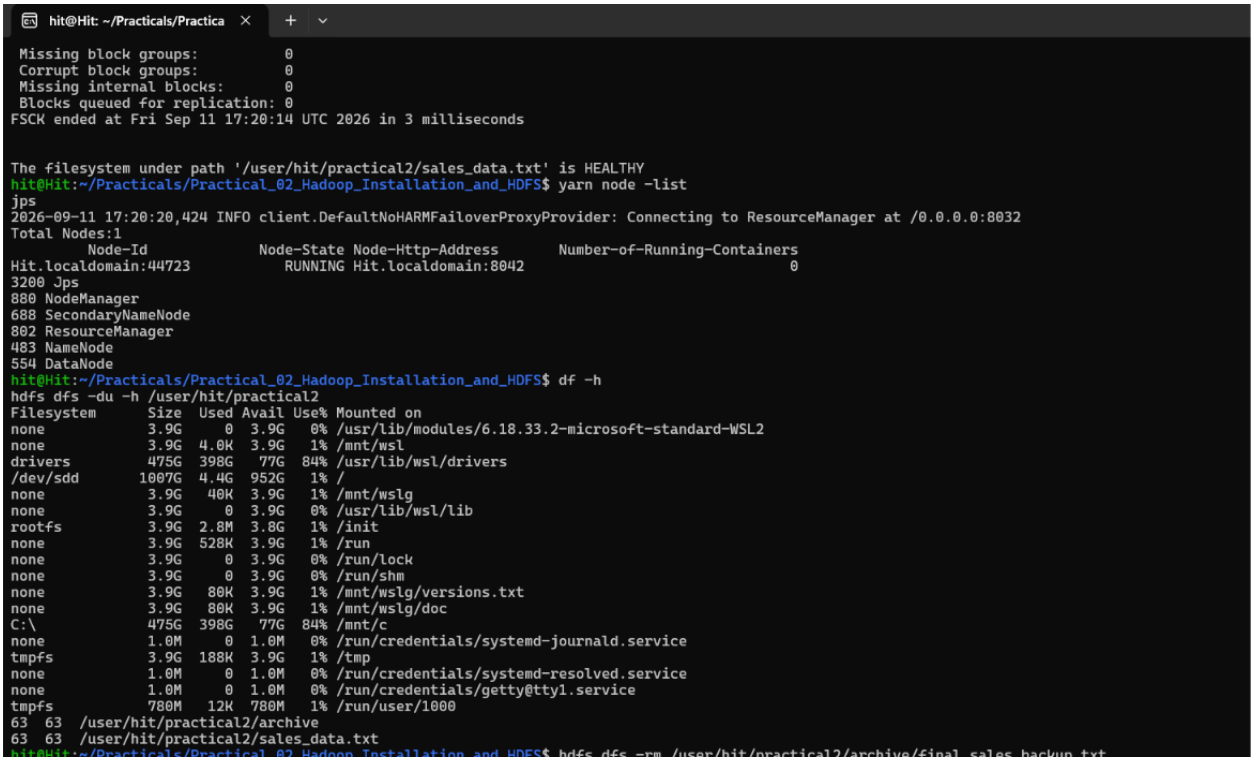
Finally, I checked the YARN node, running Hadoop processes, Linux disk usage and the size of the practical2 HDFS directory.

yarn node -list

jps

df -h

hdfs dfs -du -h /user/hit/practical2



21. Result

The Hadoop single-node setup was checked successfully, and the required HDFS operations were performed. The practical covered directory creation, file upload, listing, viewing file contents, copying, moving, renaming, downloading and deleting files. The fsck output showed the main file as HEALTHY, and the HDFS report showed one live DataNode.

22. Key Questions / Analysis

Q1. Why is Hadoop preferred for Big Data storage?

Hadoop is useful for Big Data because it is designed to store and process large amounts of data. Instead of depending on one machine, Hadoop can work with data across multiple machines in a larger cluster.

Q2. What is the role of NameNode and DataNode in HDFS?

The NameNode manages information about files and directories, such as file names and block locations. The DataNode stores the actual data blocks. In this practical, both services were part of the single-node Hadoop setup.

Q3. What is the difference between HDFS and a traditional file system?

A normal file system is generally used on a single computer or server. HDFS is designed for distributed storage, where a file can be divided into blocks and stored across DataNodes in a cluster.

Q4. What are the advantages of distributed storage?

Distributed storage can handle larger amounts of data and can use storage from multiple machines. In a larger Hadoop cluster, replication can also help improve availability. For this single-node practical, the replication factor was set to 1.

23. Conclusion

This practical gave a basic understanding of how Hadoop and HDFS are used from the command line. The commands were performed one by one, starting from checking the installation and services, then managing files in HDFS, and finally checking file health and storage details. The practical was useful for understanding how common file operations are performed in an HDFS environment.